Prime Numbers Distribution in 6D via Dimensionality Reduction

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Abstract

This document presents the dimensionality reduction concept used to visualize prime number distribution in a 6-dimensional structure projected into 2D. The method uses two prime branches and a coprime–product sieve to count distinct composites and derive the prime count formula.

1 Dimensionality Reduction Concept

The following figure illustrates the 6-side (hexahedral) structure used to represent prime number distribution in higher dimensions.

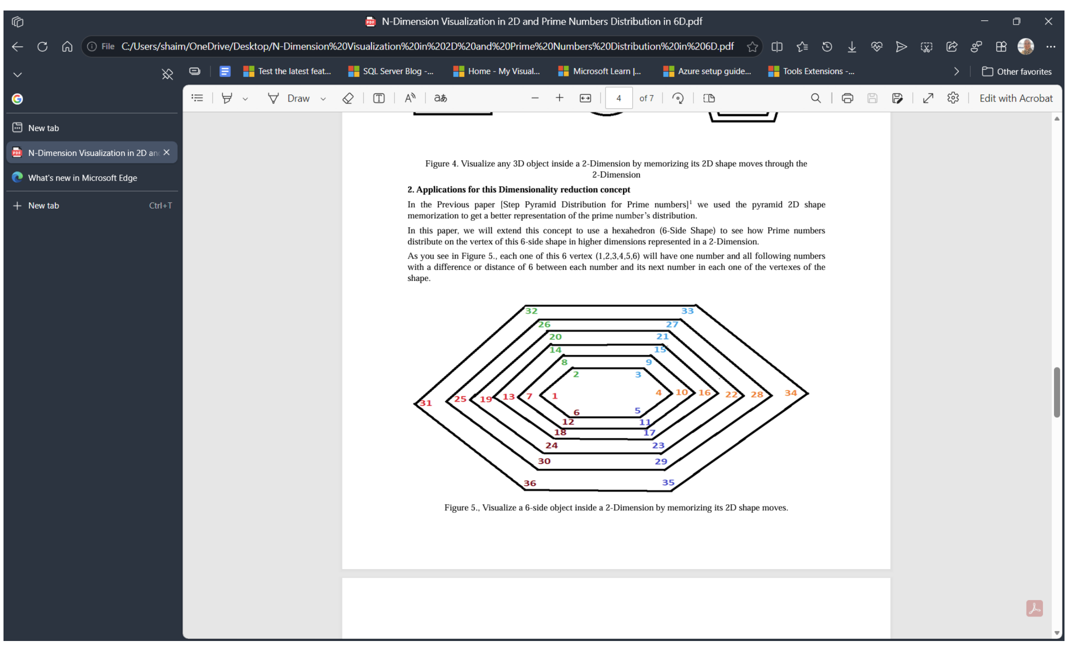

Figure 1: Visualization of a 6-side object inside 2D (Figure 5).

Each vertex of the 6-side shape contains numbers spaced by 6, forming six infinite arithmetic branches. Two of these branches contain all primes greater than 3:

$$\text{Branch } 1 = \{1, 7, 13, 19, 25, 31, \ldots\}$$

$$\text{Branch } 5 = \{5, 11, 17, 23, 29, 35, \ldots\}$$

If the shape is extended to size $6V$, each branch contains V numbers, and the two prime branches contain:

$$2V - 1$$

candidate positions.

2 Coprime–Product Matrix

The coprime–product matrix is constructed using m prime pairs. The matrix detects composite numbers by generating all coprime products across the two branches.

Below are placeholders for the two matrix images:

3 Distinct Coprime Count

The number of distinct coprime composites detected by the sieve is:

$$C_{\text{distinct}} = m(N - 1) - \sum_{i=1}^{2m-1} i$$

Using the triangular number identity:

$$\sum_{i=1}^{2m-1} i = \frac{(2m - 1)(2m)}{2}$$

Thus the closed form becomes:

$$C_{\text{distinct}} = m(N - 1) - \frac{(2m - 1)(2m)}{2}$$

4 Prime Count Formula

Since the two prime branches contain $2V - 1$ candidates, and the sieve detects all composites, the number of primes is:

$$\text{PrimeCount} = (2V - 1) - C_{\text{distinct}}$$

Substituting:

$$\text{PrimeCount} = (2V - 1) - \left[m(N - 1) - \frac{(2m - 1)(2m)}{2}\right]$$

155	341	527	713
203	377	551	725
135	297	459	621
175	325	475	625
115	253	391	529
147	273	399	525
95	209	323	437
119	221	323	425
75	165	255	345
91	169	247	325
55	121	187	253
63	117	171	225
35	77	119	161
35	65	95	125
15	33	51	69
7	13	19	25
0	0	0	0
5	11	17	23
21	39	57	75
25	55	85	115
49	91	133	175
45	99	153	207
77	143	209	275
65	143	221	299
105	195	285	375
85	187	289	391
133	247	361	475
105	231	357	483
161	299	437	575
125	275	425	575
189	351	513	675
145	319	493	667
217	403	589	775

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Figure 2: Left coprime–product matrix (distinct composite detection).

5 Variable Definitions

$$\begin{aligned}
m &= \text{number of prime pairs (columns in the coprime matrix)} \\
N &= \text{maximum index used in the coprime matrix} \\
V &= \text{length of each prime branch in the 6-side structure} \\
C_{\text{distinct}} &= \text{distinct coprime composites detected} \\
\text{PrimeCount} &= \text{number of primes in the two branches}
\end{aligned}$$

155	341	527	713	899						
203	377	551	725	899						899
135	297	459	621	783		135	297	459	621	783
175	325	475	625	775					625	
115	253	391	529	667					529	
147	273	399	525	651		147	273	399	525	651
95	209	323	437	551		95				
119	221	323	425	527				323		
75	165	255	345	435			165	255	345	435
91	169	247	325	403			169		325	
55	121	187	253	319			121			
63	117	171	225	279		63	117	171	225	279
35	77	119	161	203						
35	65	95	125	155		35				
15	33	51	69	87		15	33	51	69	87
7	13	19	25	31						
0	0	0	0	0						
5	11	17	23	29						
21	39	57	75	93		21	39	57	75	93
25	55	85	115	145		25	55		115	145
49	91	133	175	217		49	91	133	175	217
45	99	153	207	261		45	99	153	207	261
77	143	209	275	341		77		209		341
65	143	221	299	377		65	143	221	299	377
105	195	285	375	465			195	285	375	465
85	187	289	391	493		85	187	289	391	493
133	247	361	475	589		133	247	361	475	589
105	231	357	483	609		105	231	357	483	609
161	299	437	575	713		161	299	437		713
125	275	425	575	725		125	275	425	575	725
189	351	513	675	837		189	351	513	675	837
145	319	493	667	841		145	319	493	667	841
217	403	589	775	961		217	403	589	775	961

Figure 3: Right coprime–product matrix (mirrored structure).